

Activity: maximum likelihood estimation for GLMs

GLMs

We say that a distribution belongs to the *exponential dispersion model* family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$\begin{aligned} Y_i | \mathbf{x}_i &\sim \text{EDM}(\mu_i, \phi) \\ g(\mu_i) &= \mathbf{x}_i^T \beta \end{aligned}$$

For this activity, we will assume that g is the **canonical link function** – that is, $g(\mu_i) = \theta_i$ (the canonical parameter).

Furthermore, recall the following facts about EDMs:

- $\frac{d}{d\theta_i} \kappa(\theta_i) = \mu_i$
- $\frac{d^2}{d\theta_i^2} \kappa(\theta_i) = \frac{d\mu_i}{d\theta_i} = V(\mu_i)$

where $\mathbb{E}[Y_i | \mathbf{x}_i] = \mu_i$ and $\text{Var}(Y_i | \mathbf{x}_i) = \phi V(\mu_i)$.

Questions

Maximum likelihood estimation for GLMs does not, in general, have a closed form. Instead, iterative procedures like Newton's method are commonly used. Find expressions for:

- The score $U(\beta)$
- The Hessian $\mathbf{H}(\beta)$
- The Fisher information (for a single observation) $\mathcal{I}_1(\beta)$
- An *estimate* of the Fisher information

$$L(\beta | X, Y) \propto \prod_{i=1}^n f(y_i | x_i, \beta) = \prod_{i=1}^n a(y_i, \theta) \exp \left\{ \frac{y_i \theta_i - \kappa(\theta_i)}{\phi} \right\}$$

$$\propto \prod_{i=1}^n \exp \left\{ \frac{y_i \theta_i - \kappa(\theta_i)}{\phi} \right\}$$

$$\Rightarrow \ell(\beta | X, Y) = \frac{1}{\phi} \sum_{i=1}^n [y_i \theta_i - \kappa(\theta_i)]$$

$$\frac{\partial \ell}{\partial \beta} = \frac{1}{\phi} \sum_{i=1}^n \left[y_i \left(\frac{\partial \theta_i}{\partial \beta} \right) - \frac{d \kappa(\theta_i)}{d \theta_i} \frac{\partial \theta_i}{\partial \beta} \right]$$

$$\theta_i = x_i^T \beta \Rightarrow \frac{\partial \theta_i}{\partial \beta} = x_i$$

$$\frac{d \kappa(\theta_i)}{d \theta_i} = \mu_i$$

$$\text{So, } \frac{\partial \ell}{\partial \beta} = \frac{1}{\phi} \sum_{i=1}^n (y_i x_i - \mu_i x_i)$$

$$u(\beta) = \frac{1}{\phi} \sum_{i=1}^n (y_i - \mu_i) x_i$$



Questions

Maximum likelihood estimation for GLMs does not, in general, have a closed form. Instead, iterative procedures like Newton's method are commonly used. Find expressions for:

- The score $U(\beta)$
- The Hessian $\mathbf{H}(\beta)$
- The Fisher information (for a single observation) $\mathcal{I}_1(\beta)$
- An *estimate* of the Fisher information

$$u(\beta) = \frac{1}{\emptyset} \sum_{i=1}^{\hat{n}} (y_i - \mu_i) x_i$$

$$\begin{aligned} H(\beta) &= \frac{\partial u}{\partial \beta} = \frac{1}{\emptyset} \sum_{i=1}^{\hat{n}} \left(-\frac{\partial \mu_i}{\partial \beta} \right) x_i^T \\ &= -\frac{1}{\emptyset} \sum_{i=1}^{\hat{n}} \left(\frac{d\mu_i}{d\theta_i} \frac{\partial \theta_i}{\partial \beta} \right) x_i^T \\ &= -\frac{1}{\emptyset} \sum_{i=1}^{\hat{n}} \left(v(\mu_i) x_i \right) x_i^T \end{aligned}$$

$$H(\beta) = -\frac{1}{\emptyset} \sum_{i=1}^{\hat{n}} v(\mu_i) x_i x_i^T$$

Fisher info: $\mathcal{I}_1(\beta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \beta^2} \log f(y_i | x_i, \beta) \right]$ (under regularity conditions)

$$\mathcal{I}_1(\beta) = \mathbb{E} \left[\frac{1}{\emptyset} v(\mu_i) x_i x_i^T \right]$$

Estimating Fisher info: $\hat{\mathcal{I}}_1(\beta) = \frac{1}{n} \sum_{i=1}^{\hat{n}} v(\hat{\mu}_i) x_i x_i^T$
where $v(\hat{\mu}_i) = \frac{1}{\emptyset} v(g^{-1}(x_i^T \hat{\beta}))$

$$\Rightarrow \hat{\mathcal{I}}_1(\beta) = \frac{1}{\emptyset} \cdot \frac{1}{n} X^T V X$$

X = design matrix
 V = $\text{diag}(v(\hat{\mu}_i))$

Questions

Maximum likelihood estimation for GLMs does not, in general, have a closed form. Instead, iterative procedures like Newton's method are commonly used. Find expressions for:

- The score $U(\beta)$
- The Hessian $\mathbf{H}(\beta)$
- The Fisher information (for a single observation) $\mathcal{I}_1(\beta)$
- An *estimate* of the Fisher information

$$\hat{\mathcal{I}}_1(\beta) = \frac{1}{\emptyset} \cdot \frac{1}{n} X^T V X$$

Ans: $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \mathcal{I}_1^{-1}(\beta))$

So $\hat{\beta} \approx N(\beta, \frac{1}{n} \mathcal{I}_1^{-1}(\beta))$

$$\approx N(\beta, \frac{1}{n} \left(\frac{1}{\emptyset} \cdot \frac{1}{n} X^T V X \right)^{-1})$$
$$= N(\beta, \emptyset (X^T V X)^{-1})$$

Ex: linear regression ($y_i | x_i \sim N(\mu_i, \sigma^2)$)

$$\Rightarrow \hat{\beta} \approx N(\beta, \sigma^2 (X^T X)^{-1})$$

Ex: logistic regression

$$\hat{\beta} \approx N(\beta, (X^T V X)^{-1}) \quad V = \text{diag}(\hat{p}_i(1-\hat{p}_i))$$